Evaluation and Development Board Manual

for Ezairo® 7100

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CHAPTER 1
Introduction

1.1 Purpose

This manual provides detailed information about the configuration and use of the Ezairo® 7100 Evaluation and Development Board. The Evaluation and Development Board consists of two printed circuit boards: a motherboard and daughterboard that are designed for use with the software tools included with the Evaluation and Development Kit to evaluate the performance and capabilities of the Ezairo 7100 audio processing system.

1.2 Intended Audience

This manual is for engineers, developers, and anyone else who requires technical information about the Evaluation and Development Board.

1.3 Conventions

The following conventions are used in this manual to signify particular types of information:

`monospace font` Connector and pin names.

1.4 Manual Organization

The *Evaluation and Development Board Manual for Ezairo 7100* contains the following chapters and appendices:

- **Chapter 1: Introduction**, describes the purpose of this manual, describes the target reader, explains how the book is organized, and provides a list of suggested reading for more information.
- **Chapter 2: Overview**, provides an overview of the Evaluation and Development Board described in this manual.
- **Chapter 3: Evaluation and Development Board**, provides the details of the Evaluation and Development Board. The chapter is divided into the following topics:
  - Development Board Setup
  - Development Board Design
  - Power Supply
  - Level Translators
  - LED Circuitry
  - Ezairo 7100 Chip
  - Audio Interfaces
  - EEPROM
  - Low-speed A/D
  - I^2C Port
  - Digital Input/Output
  - User Configurations
  - Prototyping Area and Test Points

- **Appendix A: Connectors**, provides a complete list of the connectors and jumpers on the Evaluation and Development Board. The appendix is divided into the following sections:
  - Configuration header jumpers
  - Header descriptions
  - Switch descriptions and configurations
  - Connector descriptions
  - Potentiometer descriptions

- **Appendix B: Motherboard Schematics**, contains the motherboard schematics for the Evaluation and Development Board.
- **Appendix C: Daughterboard Schematics**, contains the daughterboard schematics for the Evaluation and Development Board.

1.5 Further Reading

For more information, refer to the following documents:

- *Promira™ Serial Platform Quick Start Guide* available from http://www.totalphase.com/support/articles/203908616-Promira-Serial-Platform-Quick-Start-Guide
- *Promira Serial Platform User Manual* available from http://www.totalphase.com/support/articles/204227283-Promira-Serial-Platform-User-Manual
- *Communication Accelerator Adaptor Manual*, which is installed with the Communication Accelerator Adaptor (CAA) software
- *Integrated Development Environment User's Guide*
- *Introduction to Ezairo 7100 Programming*
- *CFX DSP Architecture Manual*
- *Firmware Reference Manual for Ezairo 7100*
- *Hardware Reference Manual for Ezairo 7100*
- *Ezairo 7100 Errata*
- *HEAR Configurable Accelerator Reference Manual for Ezairo 7100*
- *SK4 ELF Toolchain Reference for Ezairo 7100*
- Ezairo 7100 Datasheet
- Analog Devices ADG-3304 Datasheet, Rev. B, December 2005
- Texas Instruments TPS60302 Datasheet no. SLVS302A, March 2001

CHAPTER 2
Overview

2.1 Introduction

The Ezairo 7100 Evaluation and Development board consists of a motherboard and daughterboard that are used for evaluating Ezairo 7100 and for application development. The motherboard provides access to all input and output connections via 0.1" standard headers or RCA connectors. One 6-pin DIN socket (J2) is available on the motherboard to connect to the Promira Serial Platform that provides communication to the board from a host PC. Alternatively, you can communicate using one of two 6-pin DIN sockets (`J1` and `J2`) from the motherboard to connect to the Communication Accelerator Adaptor (CAA). These connectors can also be used for connecting to other Ezairo 5900, Ezairo 6200 or Ezairo 7100 platforms. For more information about Promira, see the *Promira Serial Platform User Manual*. For more information about the CAA, see the *Communication Accelerator Adaptor Manual*, which is installed with the CAA software.

The daughterboard provides access to the VDDO2 digital I/O bank and provides additional circuitry for assisting application developers. All voltage level translators, LEDs, switches and volume controls are provided on the daughterboard.

2.2 Evaluation and Development Board Features

The Evaluation and Development Board enables developers to evaluate the performance and capabilities of the Ezairo 7100 audio processing system in addition to developing, demonstrating and debugging applications.

Key features of the motherboard include:

- An onboard 512K EEPROM and DIP socket for other EEPROMs
- Standard audio connectors (RCA, 0.1" headers)
- Access to all Ezairo 7100 peripherals via standard 0.1" headers
- Easy interface to the Total Phase™ Promira Serial Platform or ON Semiconductor Communication Accelerator Adaptor for communication from a host PC to Ezairo 7100
- Application and output filter prototyping areas

Key features of the daughterboard include:

- Level translators for digital input/output voltage translation
- Momentary push button switches and LEDs for selecting and monitoring DIOs
- A potentiometer for volume control applications using the LSADs
- An application prototyping area

CHAPTER 3
Evaluation and Development Board

3.1 Development Board Setup

This section is an overview of how to configure the Evaluation and Development Board. Details of the development board configuration are discussed later in this manual.

Figure 1 represents an overview of the board setup.

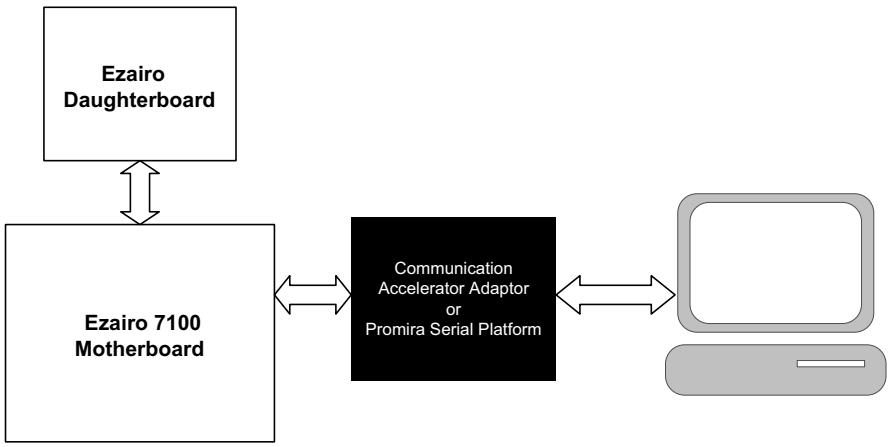


Figure 1. Evaluation and Development Board Setup

The Ezairo 7100 chip is packaged in a 100 pin TQFP package that is used on the Evaluation and Development Board.

The Ezairo 7100 Evaluation and Development Board can be powered from different sources:

- The Promira Serial Platform or the Communication Accelerator Adaptor (CAA)
- An external power source
- A coin cell battery onboard.

See Section 3.3, “Power Supply” on page 10 for details about each of these power sources.

CAUTION: Do not connect an external power supply and a battery at the same time. Connecting both might damage the board, the battery, or both the board and battery.

3.2 Development Board Design

The following sections detail the various sub-circuits of the Ezairo 7100 Evaluation and Development Board; both boards are outlined. The block diagram in Figure 2 shows the locations of the various circuit sections for the motherboard.

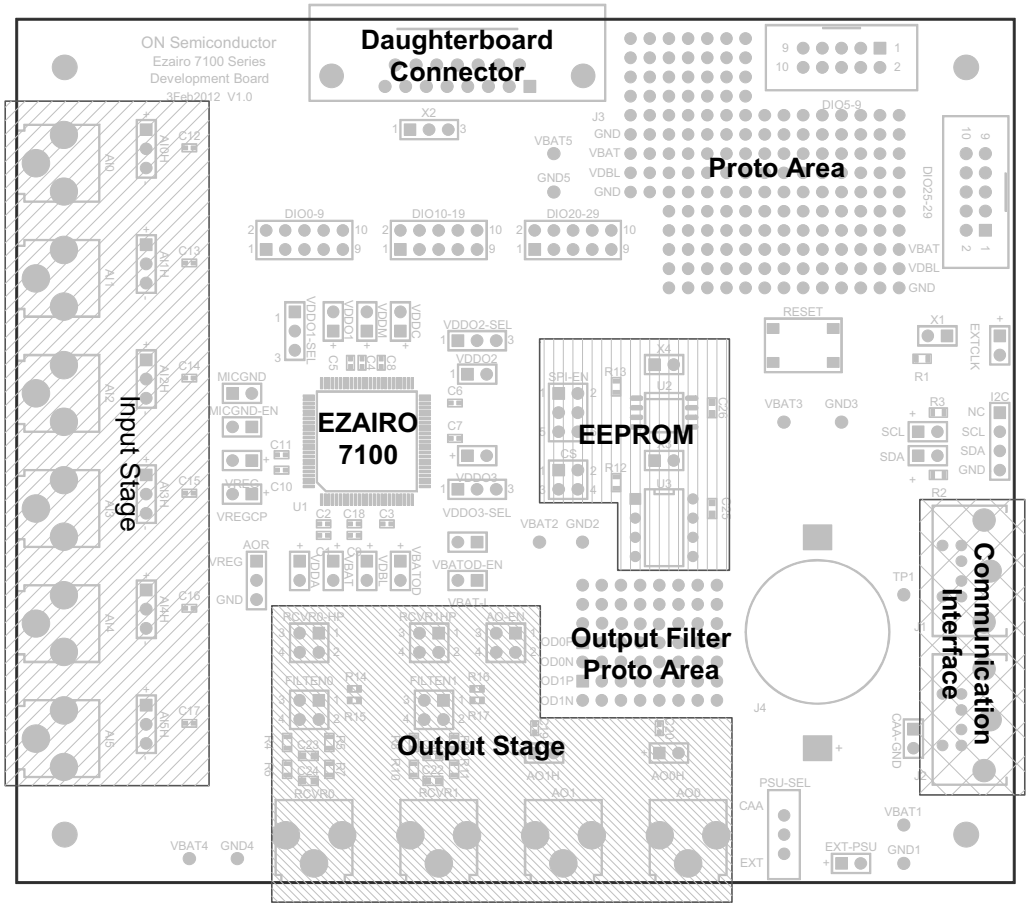


Figure 2. Motherboard Circuit Location Block Diagram

The motherboard connects directly to the daughterboard via a 15-pin DSUB connector. A 15-pin plug DSUB connector is provided on the motherboard whereas the daughterboard uses a 15-pin receptacle DSUB connector. The two boards connect directly to each other via these connectors (i.e., no cable is required).

A block diagram showing all the daughterboard circuit sections is shown in Figure 3.

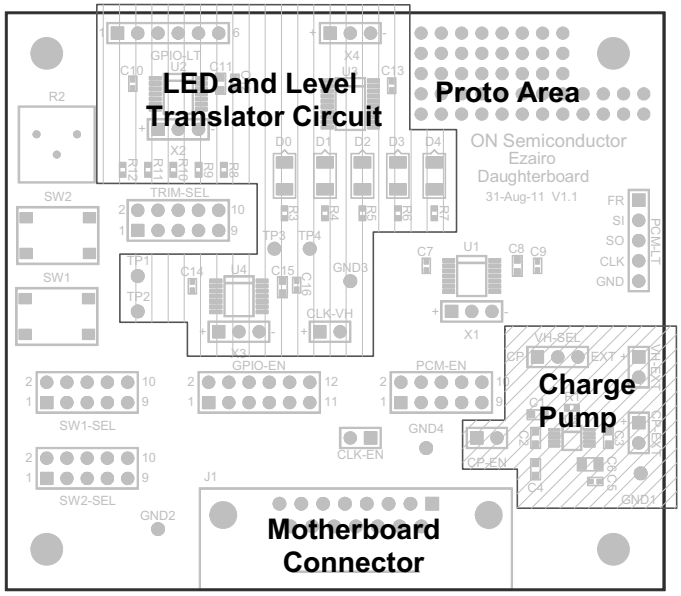


Figure 3. Daughterboard Circuit Location Block Diagram

3.3 Power Supply

The Evaluation and Development Platform can be powered by one of the following:

- The Promira Serial Platform via `J2` or the CAA via `J1` or `J2`
- An external supply connected to the `EXT-PSU` header
- A battery inserted into the battery holder (`J4`) located on the motherboard.

Use the `PSU-SEL` switch to select a power supply option as shown in Table 1.

Table 1. Power Supply Selection

Power Source	PSU-SEL
Promira Serial Platform	CAA
Communication Accelerator Adapter	CAA
Battery (No External Supply)	EXT
External Supply (No Battery)	EXT

CAUTION: When using either an external power source or battery, ensure that these power sources are not connected at the same time. The `EXT` position on the `PSU-SEL` switch is connected to both the `EXT-PSU` header and battery input. If both power sources are connected, the board and battery might be damaged.

The TPS60302 is a high efficiency charge pump located on the daughterboard that provides a regulated 3.3 V output typically used for the level translator and LED circuits when the VH-SEL header is configured to CP. The charge pump provides a maximum current of 20 mA with an input of 1 V.

CAUTION: When connecting the charge pump to other devices, ensure that the current does not exceed 20 mA or the charge pump will lose regulation.

The CP-EN header on the daughterboard enables or disables the input voltage to the charge pump. VBAT provides the input voltage to the charge pump from the motherboard when the CP-EN header is enabled. Alternatively, to use an external supply, connect the power source to the CP-EXT header. Table 2 shows the CP-EN header configurations for the charge pump.

Table 2. Charge Pump Configurations

Charge Pump Input	CP-EN
Motherboard VBAT Supply	shorted
External Supply	open
Disabled	open

Either the charge pump output (3.3 V) or an external supply can provide the high voltage reference (Vh) for the level translators and LED buffer-driver. The VH-EXT header is provided for connecting an external power source for the Vh voltage of the level translators and LED circuit. When an external source is used, set the VH-SEL header to EXT. Refer to Table 3 for the VH-SEL header configurations.

Table 3. VH Supply Selection

Condition	VH-SEL
Charge Pump	short middle pin to CP
External Supply	short middle pin to EXT

NOTE: An error in the silkscreen has reversed the VH-SEL header labels. To select the charge pump, short the middle pin to EXT; for an external supply, short the middle pin to CP.

Ensure that the recommended voltage levels are not exceeded when using external power sources for the motherboard and daughterboard. Table 4 indicates the maximum/minimum voltage levels for each of the external power supplies. Note that the Vh minimum voltage is based on the minimum voltage of the LED buffer-driver, and the Vh maximum voltage is based on the maximum voltage for the level translators. Further information regarding the level translators can be found in Section 3.4, “Level Translators” on page 12.

Table 4. Minimum/Maximum External Regulated Voltages

Power Supply	Header	Input Voltage		
		Minimum	Typical	Maximum
Motherboard (Ezairo 7100)	EXT-PSU	1.05 V	1.25 V	2.0 V
Daughterboard (Charge Pump Input)	CP-EXT	0.9 V	1.0 V	1.8 V
Daughterboard (Vh)	VH-EXT	2.4 V	3.3 V	5.5 V

3.4 LEVEL TRANSLATORS

The daughterboard has level translators for the VDDO2 DIO bank signals of Ezairo 7100, including the clock signal. The level translators facilitate interfacing to external devices that operate at a higher voltage than Ezairo 7100. The level translators are also used in conjunction with the GPIO LED circuitry described in Section 3.5, “LED Circuitry” on page 13.

The level translation circuitry consists of components U1, U2, U4 and the GPIO-LT header. Signals are translated from the low Vio voltage reference to higher Vh voltage provided by the charge pump output or by an external supply. The VDDO2 voltage from Ezairo 7100 provides the reference voltage for the Vio supply. The VDDO2 voltage is configured by the VDDO2-SEL header located on the motherboard to either VBAT or VDBL, as shown in Table 8 on page 14.

The ADG3304 level translators are bi-directional. From the manufacturer specifications:

- VCCY must be at least 0.3 V higher than VCCA for VCCA = 1.15 V
- VCCY must be at least 0.4 V higher than VCCA for VCCA = 1.2 V - 5.5 V
- VOHA is equal to VCCY - 0.4 @ 20 µA
- VCCA minimum is 1.15 V (low voltage threshold)

To disable the level translators, configure the headers X1, X2, and X3 on the daughterboard; refer to Table 5. The default configuration is enabled. Specific configurations are outlined in Section 3.12, “User Configurations” on page 19 and in Appendix A, “Connectors” on page 21.

Table 5. Level Translator Header Configurations

Daughterboard Header	Level Translator	Enabled	Disabled
X1	PCM (DIO16 - DIO19)	short middle pin to "+"	short middle pin to "-"
X2	DIO10 - DIO13	short middle pin to "+"	short middle pin to "-"
X3	DIO14 and Ezairo 7100 Clock	short middle pin to "+"	short middle pin to "-"

Short headers GPIO-EN and PCM-EN to level translate these signals. Refer to Table 6 for details.

Table 6. GPIO-EN and PCM-EN Configurations

Header	Enabled	Disabled
GPIO-EN	short pins 1 to 2	open pins 1 to 2
	short pins 3 to 4	open pins 3 to 4
	short pins 5 to 6	open pins 5 to 6
	short pins 7 to 8	open pins 7 to 8
	short pins 9 to 10	open pins 9 to 10
PCM-EN	short pins 1 to 2	open pins 1 to 2
	short pins 3 to 4	open pins 3 to 4
	short pins 5 to 6	open pins 5 to 6
	short pins 7 to 8	open pins 7 to 8

For information about enabling specific DIOs for level translation, see Table 22 on page 21 and Figure 8 on page 34.

3.5 LED Circuitry

The LED circuitry consists of the level translator circuitry and an octal buffer (U3). Header X4 is provided to enable or disable the buffer. To use the LED circuitry, enable both the buffer and level translators as well as the GPIO-EN header. The default configuration is enabled; refer to Table 7 for details.

Table 7. LED Buffer-Driver Header Configuration

Enabled	Disabled
short middle pin to "-"	short middle pin to "+"

3.6 Ezairo 7100 Chip

The Ezairo 7100 VBAT main system supply can be provided by the Promira's 5 V ±5% regulated supply, or the CAA's regulated 1.25 V supply, a battery inserted in the battery holder (J4), or by an external power source. For more information, refer to Section 3.3, "Power Supply" on page 10.

The VBAT-I header located on the motherboard is provided to measure the current consumption of Ezairo 7100. To properly assess Ezairo 7100 current consumption, disconnect the daughterboard. For normal operation, short the VBAT-I header. When using an external supply, ensure that the recommended voltage level for Ezairo 7100 is not exceeded; refer to Table 4 on page 12 for minimum and maximum voltages.

Headers are provided for each of the regulated Ezairo 7100 voltages for additional capacitance and/or for measurements. Ezairo 7100 has 30 digital I/Os that are mapped into three power domains: VDDO1, VDDO2 and VDDO3. The headers VDDO1-SEL, VDDO2-SEL and VDDO3-SEL (refer to Table 8) configure the I/O voltages for each of these power domains to either VBAT or to VDBL (a regulated supply from Ezairo 7100). The VDDO1 I/O voltage is connected to the onboard SPI EEPROM and is typically set to VDBL. The VDDO2 I/O voltage connects to the 15-pin DSUB connector that is used as the low voltage reference (Vio) for the level translators located on the daughterboard. The VDDO3 I/O voltage is also used by Ezairo 7100 for its debug port interface, reset pin and external clock input.

Table 8. I/O Voltage Configurations

Header	VBAT	VDBL
VDDO1-SEL	short pins 1 to 2	short pins 2 to 3
VDDO2-SEL	short pins 1 to 2	short pins 2 to 3
VDDO3-SEL	short pins 1 to 2	short pins 2 to 3

The default configuration for the VBATOD-EN header connects the VBATOD and VBATOD1 pins to VBAT. For high power applications, you might need to add an RC filter to the VBAT pin on Ezairo 7100 to reduce AC ripple caused by driving low impedance loads (refer to Table 9). Additional information regarding high power mode is available in Section 3.7.2, “Output Stage” on page 15.

Table 9. Receiver Supply Configurations

Function	VBAT-I
Low Power Mode	short
High Power Mode	add resistor if necessary

The RC filter consists of the resistor placed across the VBAT-I header and capacitor C1 (1 µF). To increase the capacitance, add a capacitor to header VBAT.

3.7 Audio Interfaces

3.7.1 Input Stage

The audio input interface consists of the following connectors located on the motherboard:

- Six standard monaural RCA connectors: AI0, AI1, AI2, AI3, AI4 and AI5.
- Six standard 0.1" headers: AI0H, AI1H, AI2H, AI3H, AI4H and AI5H.

The input ground reference for the above input connectors is normally referenced internally to Ezairo 7100 GND_MIC. The board provides two headers to change the input ground reference to either GND_MIC or to GNDA. Headers MICGND-EN and MICGND configure the input ground reference; refer to Table 10.

Table 10. Input Ground Reference Configuration

Function	Header	Configuration
Input Ground to GND_MIC	MICGND-EN	short
	MICGND	open
Input Ground to GNDA	MICGND-EN	open
	MICGND	short

For electrical testing of Ezairo 7100, we recommend configuring the input ground to GNDA when the test equipment provides a single-ended (unbalanced) output for supplying the audio input signal. For test equipment that provides differential outputs (balanced), you can connect the input ground to GND_MIC. We recommend performing electro-acoustic testing (with the microphone and receiver attached to the PCB) with the input ground connected to GND_MIC.

3.7.2 Output Stage

Two digital outputs (RCVR0, RCVR1) and two analog outputs (AO0, AO1) are provided on the motherboard. The audio output interface consists of the following connectors located on the motherboard:

- Four standard monaural RCA connectors: RCVR0, RCVR1, AO0 and AO1
- Standard 0.1" headers: FILTEN0, FILTEN1, RCVR0HP, RCVR1HP, AO-EN, AO0H and AO1H
- Two RC filter networks for use with the direct digital outputs when running electrical measurements

Outputs for the direct digital outputs are available on headers FILTEN0 and FILTEN1, and the RCA connectors, RCVR0 and RCVR1. Separate RC filter networks are provided to attenuate out-of-band noise from the direct digital outputs when connected to high impedance audio measurement equipment. The onboard RC filters are enabled by the FILTEN0 and FILTEN1 headers as described in Table 11.

Table 11. On-Board RC Filter Configuration

Function	FILTEN0	FILTEN1
Disabled	open pins 1 to 2	open pins 1 to 2
	open pins 3 to 4	open pins 3 to 4
Enabled	short pins 1 to 2	short pins 1 to 2
	short pins 3 to 4	short pins 3 to 4

To connect receivers or loud speakers (without the RC filter), use headers FILTEN0 and FILTEN1 pins 1 and 3.

A prototyping area allows you to use customer-specific filters with the digital drive output. Four plated thru-holes—OD0P, OD0N, OD1P and OD1N—are located close to the edge of the output filter prototyping area for easy connection. To use these test points, you must install 0402 size 0 Ω resistors on R14, R15, R16 and R17.

The direct digital outputs can be configured to operate in high power mode. In high power mode, both parts of the output stage are arranged in parallel to effectively reduce the output impedance by half, and achieve higher maximum output. To use the high power mode, configure the RCVR0HP and RCVR1HP headers as explained in Table 12. You might need an RC filter to reduce the AC ripple caused by driving a low impedance load; refer to Section 3.6, “Ezairo 7100 Chip” on page 13 for more information.

Table 12. High Power Mode Configuration

Function	RCVR0HP	RCVR1HP
Disabled	open pins 1 to 2	open pins 1 to 2
	open pins 3 to 4	open pins 3 to 4
Enabled	short pins 1 to 2	short pins 1 to 2
	short pins 3 to 4	short pins 3 to 4

The header AO-EN connects the Ezairo 7100 analog output signals to the analog output headers (AO0H or AO1H) and RCA jacks (AO0 or AO1).

Table 13 lists the AO-EN header configurations.

Table 13. Analog Output Enable/Disable Configuration

Function	AO-EN
AO0 and AO1 not connected to onboard connectors	open pins 1 to 2
	open pins 3 to 4
AO0 and AO1 connected to onboard connectors	short pins 1 to 2
	short pins 3 to 4

The Ezairo 7100 analog output stage is referenced to AOR where header AOR configures this reference to either VREG or AGND. The header configuration of AOR also sets the ground reference for the analog output connectors. The AOR ground is typically set to AGND; refer to Table 14.

Table 14. Analog Output AOR Configuration

Function	AOR
VREG	short pins 1 to 2
AGND	short pins 2 to 3

3.8 EEPROM

The motherboard has a 512K EEPROM (U2) along with an 8-pin DIP socket (U3) for connecting to another EEPROM. To disable the supply voltage for each EEPROM, use headers X4 and X3 respectively. The VDDO1 I/O voltage supplies power for the EEPROMs.

Refer to Table 15 for configuring the board to use the onboard EEPROM (U2).

Table 15. EEPROM Header Configuration (U2)

Header	Configuration
SPI-EN	short pins 1 to 2
	short pins 3 to 4
	short pins 5 to 6
X4	short
CS	short pins 1 to 2

Similarly, refer to Table 16 for configuring the board to use an EEPROM installed in the 8-pin DIP socket (U3).

Table 16. EEPROM Header Configuration (U3)

Header	Configuration
SPI-EN	short pins 1 to 2
	short pins 3 to 4
	short pins 5 to 6
X3	short
CS	short pins 3 to 4

3.9 Low-Speed A/D (LSAD)

The LSAD inputs are multiplexed with the first 6 pads in each bank of 10 DIOs. LSADs are available on DIOs 0 to 5, 10 to 15, and 20 to 25. Headers for the DIO banks can be used to access the LSAD inputs. For more information about the DIO header banks, see Section 3.11, “Digital Input/Output (DIO)” on page 18.

NOTE: The input read for an individual LSAD is undefined if multiple DIOs are configured as the source for that LSAD. For example, if both DIO0 and DIO10 are defined as LSAD inputs, the output for LSAD 0 is undefined.

LSAD signals on DIO10 to DIO14 are routed via the 15-pin DSUB connector on the motherboard to the daughterboard. A potentiometer (R2) on the daughterboard provides a voltage reference between VBAT and GND for volume control applications. The header TRIM-SEL located on the daughterboard selects the LSAD to connect to the potentiometer input. Table 17 shows the TRIM-SEL pin-out.

Table 17. LSAD Selection

Configuration	TRIM-SEL
DIO10	short pins 1 to 2
DIO11	short pins 3 to 4
DIO12	short pins 5 to 6
DIO13	short pins 7 to 8
DIO14	short pins 9 to 10

When using the LSAD, disable the level translators by opening the GPIO-EN header or disabling the level translators as described in Section 3.12, “User Configurations” on page 19. This prevents the LSAD from reading erroneous voltages.

3.10 I^2C Port

The Promira or CAA are typically used to communicate with Ezairo 7100 using an I^2C communication protocol. The 6-pin DIN header (J2) on the motherboard connects Ezairo 7100 to Promira. Alternatively, you can connect the I^2C header on the motherboard to an external I^2C host.

If you have a CAA, two 6-pin DIN connectors (J1 or J2) on the motherboard connect Ezairo 7100 to it. Alternatively, the motherboard has a 4-pin header (I2C) that can be used to connect to the I^2C interface.

Headers SDA and SCL allow you to add pull-up resistors to the SDA and SCL signals. You can also add 0603 size resistors to the R2 and R3 footprints provided on the motherboard.

The I^2C connection carries I^2C bus signals, as well as power, ground, and the VDDO3 signal (J1 only) where VDDO3 is configurable to either VBAT or VDBL (refer to Table 8 on page 14). Table 18 shows the pin number and associated signal of the primary I/O connectors: J1 and J2. Use J1 if you are using the Vsense mode of the CAA (this is the default). Use J2 to connect to other communication hardware.

Table 18. DIN Connector (J1 and J2) Pin-Out on the Motherboard

Pin Number	J1 Pin Description	J2 Pin Description
1	Supply Voltage	
2	System Ground	
3	SCL (I^2C Clock)	
4	SDA (I^2C Data)	
6	VDDO3 (VBAT or VDBL) (Sense Voltage)	No Connect
6	No Connect	

3.11 DIGITAL INPUT/OUTPUT (DIO)

Ezairo 7100 contains 30 digital I/O (DIO) signals that are split into 10 signals per VDDO supply voltage domain. The three DIO voltage domains are VDDO1, VDDO2 and VDDO3 where the input voltage can be VBAT or VDBL as outlined in Section 3.6, “Ezairo 7100 Chip” on page 13 (refer to Table 8 on page 14).

The DIO signals on Ezairo 7100 are multiplexed with several interfaces, including:

- GPIO (controlled by the CFX DSP or the ARM® Cortex-M3 core)
- SPI (x2)
- PCM (x2)
- I^2C
- UART
- LSAD inputs (x6)
- Output clocks
- Other digital inputs and outputs

For more information about the DIO multiplexed signals, refer to the *Hardware Reference Manual for Ezairo 7100*.

The motherboard provides access to any of the DIOs or their multiplexed signals via the `DIO0-9`, `DIO10-19` and the `DIO20-29` headers. Two 10-pin protected headers are provided on the motherboard to provide backward compatibility with other evaluation and development boards. The `DIO5-9` header is connected to signals DIO5 to DIO9, and the `DIO25-29` header is connected to signals DIO25 to DIO29. Both of the protected headers have the even numbered pins connected to ground.

The `GPIO-LT` header located on the daughterboard provides access to the DIO10 to DIO14 level translated signals. The `PCM-LT` header located on the daughterboard provides access to the DIO16 to DIO19 level translated signals.

On the motherboard, header `X2` configures pin 6 on the 15-pin DSUB connector (`J3`) to DIO15 or ground. The Ezairo series daughterboard connects pin 6 to ground; thus, if connecting an Ezairo series daughterboard, you must ensure that header `X2` on the motherboard is configured to ground (short pins 2 to 3).

CAUTION: Ensure that `X2` is shorted between pins 2 and 3 before connecting the Ezairo 7100 Evaluation and Development Board to an Ezairo series daughterboard, or you might cause a short circuit.

The daughterboard has two momentary switches that you can configure via the SW1-SEL and SW2-SEL headers to use any of DIOs 10 to 14. These momentary switches connect the selected DIO to ground. Table 19 shows the switch header configurations for both SW1-SEL and SW2-SEL.

Table 19. SW1 and SW2 DIO Configuration

SW1-SEL/SW2-SEL	Configuration
DIO10	short pins 1 to 2
DIO11	short pins 3 to 4
DIO12	short pins 5 to 6
DIO13	short pins 7 to 8
DIO14	short pins 9 to 10

The LED circuit provides visual monitoring of the DIOs; refer to Section 3.5, “LED Circuitry” on page 13 for further information.

IMPORTANT: To avoid unintended side effects, ensure that only one DIO is selected for each of the `SW1-SEL`, and `SW2-SEL` and `TRIM-SEL` headers!

The daughterboard provides access to a set of level translated signals via header `PCM-LT`, which Ezairo 5900 and Ezairo 6200 historically used for PCM data. The Ezairo 7100 motherboard connects these level translated signals to DIOs 16 to 19, which you can configure for either of the two PCM interfaces supported by Ezairo 7100, in addition to configuring any of the other DIO supported interfaces. The PCM level translation circuitry residing on the daughterboard consists of headers `PCM-EN`, `X1`, `PCM-LT` and `IC U1`. To disable the circuit, short the middle pin to the "-" pin on header `X1`, or open all jumpers on the `PCM-EN` header.

The `PCM-EN` header pin-out is shown in Table 20.

Table 20. PCM-EN Header Configuration

Configuration	PCM-EN
PCM Frame	Short pins 1 to 2
PCM Serial Data Input	Short pins 3 to 4
PCM Serial Data Output	Short pins 5 to 6
PCM Clock	Short pins 7 to 8
Ground	Pins 9 and 10

3.12 User Configurations

Certain user configurations using either the DIOs or the multiplexed signals require adjustments to various headers on the daughterboard as described below.

LSAD Interface:

- Configure the Ezairo 7100 DIO10, DIO11, DIO12, DIO13 or DIO14 signals as an LSAD.
- Disable level translators (`U2` and `U4`) by configuring headers `X2` and `X3` to "-". Disconnect all jumpers from the `GPIO-EN` header.
- Disable the LED circuit; short header `X4` to "+".
- Select one DIO (configured as an LSAD) for the onboard potentiometer by configuring the `TRIM-SEL` header.
- Ensure that the selected LSAD is not used for `SW1-SEL` or `SW2-SEL`.

PCM Interface:

- Configure the Ezairo 7100 DIOs as one of the PCM interfaces:
  - Signal DIO19 as a digital input and source for PCM-FR (if operating as a PCM slave) or as PCM-FR (if operating as a PCM master)
  - Signal DIO17 as a digital input and as the source for PCM-SERI
  - Signal DIO18 as PCM-SERO
  - Signal DIO16 as a divided clock output (if Ezairo 7100 is providing the clock), or as a digital input (if the clock is provided externally, and as the source for PCM-CLK)

- Jumper pins on the PCM-EN header (refer to Table 20 on page 19).
- Enable the level translator circuit by setting header X1 to "+".
- Ensure that the correct voltage is applied to the high voltage input to the level translators (Vh).
  - Charge Pump (VBAT Input):
    - CP-EN shorted
    - Short VH-SEL to CP (due to an error in the silkscreen, short VH-SEL to EXT)
    - Ensure that Vh is 3.3 V

  - External Supply:
    - CP-EN open
    - Short VH-SEL to EXT (due to an error in the silkscreen, short VH-SEL to CP)
    - Apply a power source to the VH-EXT header
    - Ensure that the voltage is between 2 and 5.5 V

SW1/SW2 Configuration:

- Select one DIO (configured as a GPIO) for SW1 and SW2 by configuring the SW1-SEL and SW2-SEL headers; refer to Table 19 on page 19 for configuration information.
- Ensure a different GPIO is selected for each switch.
- Ensure the selected GPIOs are not selected on the TRIM-SEL header.

3.13 Prototyping Area and Test Points

Prototyping areas that accommodate 0.1" standard headers are provided on the motherboard and daughterboard for user development. Two prototyping areas are provided on the motherboard: one general-purpose area and another for output stage filters. The general-purpose prototyping area provides readily accessible VBAT, VDBL and GND test points. The output stage prototyping area provides easy access to test points OD0P, OD0N, OD1P and OD1N if 0 Ω 0402 resistor to R14, R15, R16 and R17 (as described in Section 3.7, “Audio Interfaces” on page 14).

You can also access signals on various headers on both boards as described throughout this document.

CHAPTER A
Connectors

A.1 Overview

This appendix lists all connectors on the Evaluation and Development Board. The following sections provide descriptions for:

- Jumpers and their possible configurations
- Headers
- Switches and their possible configurations
- Connectors
- Potentiometers

A.2 Configuration Header Jumpers

Table 21. Motherboard Jumper Descriptions

Designator	Description
X1	External oscillator 50 Ω impedance enable
X2	15-pin DSUB pin 6 configuration, DIO15 or ground
CAA-GND	Disconnects the CAA ground from the system ground
VDDO1-SEL	Ezairo 7100 VDDO1 voltage selection
VDDO2-SEL	Ezairo 7100 VDDO2 voltage selection
VDDO3-SEL	Ezairo 7100 VDDO3 voltage selection
VBAT-I	Ezairo 7100 VBAT current header and power enable
VBATOD-EN	Ezairo 7100 VBATOD and VBATOD1 power enable (output stage)
MICGND	Input ground connection to AGND
MICGND-EN	Disconnects GND_MIC from the input ground
AOR	Ezairo 7100 AOR reference selection
AO-EN	Ezairo 7100 analog output connectors enable/disable
RCVR0HP	Ezairo 7100 OD0 high power mode enable/disable
RCVR1HP	Ezairo 7100 OD1 high power mode enable/disable
FILTEN0	Enable OD0 onboard RC output filter
FILTEN1	Enable OD1 onboard RC output filter
SPI-EN	EEPROM SPI enable
CS	EEPROM chip select
X3	DIP socket EEPROM power enable
X4	Onboard 512K EEPROM power enable

Table 22. Daughterboard Jumper Descriptions

Designator	Description
CP-EN	VBAT connected to input of charge pump
VH-SEL	Level translator high voltage input selection
SW1-SEL	SW1 input selection
SW2-SEL	SW2 input selection

Table 22. Daughterboard Jumper Descriptions (Continued)

Designator	Description
TRIM-SEL	Potentiometer (R2) input selection
PCM-EN	Connect DIOs configured as a PCM or other interface to input of level translators
GPIO-EN	Connect DIOs configured as GPIOs to input of level translators
CLK-EN	Enable clock input to level translators from any Ezairo series EXT_CLK pin
X1	PCM level translator enable/disable
X2	DIO10 to DIO13 level translator enable/disable
X3	DIO14 and CLK level translator enable/disable
X4	LED buffer-driver enable/disable

Table 23. Motherboard Power Supply Jumper Configuration

Designator	Value Symbol	Pin Configuration
CAA-GND	CAA_GND_NO_CONNECT	open
	CAA_GND_CONNECT	short
VDDO1-SEL	VDDO1_VBAT	short pins 1 to 2
	VDDO1_VDBL	short pins 2 to 3
VDDO2-SEL	VDDO2_VBAT	short pins 1 to 2
	VDDO2_VDBL	short pins 2 to 3
VDDO3-SEL	VDDO3_VBAT	short pins 1 to 2
	VDDO3_VDBL	short pins 2 to 3
VBAT-I	EZAIRO_VBAT_DISABLE	open
	EZAIRO_VBAT_ENABLE	short
	EZAIRO_VBAT_ENABLE_HIPWR_MODE	add resistor
VBATOD-EN	EZAIRO_VBATOD_DISABLE	open
	EZAIRO_VBATOD_ENABLE	short
MICGND	INPUTGND_TO_GND_MIC	open
	INPUTGND_TO_AGND	short
MICGND-EN	INPUTGND_TO_GND_MIC	short
	INPUTGND_TO_AGND	open

Table 24. Motherboard Digital Signals Jumper Configurations

Designator	Value Symbol	Pin Configuration
X1	EXTOSC_50OHM_DISABLE	open
	EXTOSC_50OHM_ENABLE	short
X2	J3_PIN6_DIO15_SELECT	short pins 1 to 2
	J3_PIN6_GND_SELECT	short pins 2 to 3
X3	DIP_EEPROM_POWER_DISABLE	open
	DIP_EEPROM_POWER_ENABLE	short

Table 24. Motherboard Digital Signals Jumper Configurations (Continued)

Designator	Value Symbol	Pin Configuration
X4	ONBOARD_EEPROM_POWER_DISABLE	open
	ONBOARD_EEPROM_POWER_ENABLE	short
SPI-EN	ONBOARD_EEPROM_CLK_DISABLE	open pins 1 to 2
	ONBOARD_EEPROM_CLK_ENABLE	short pins 1 to 2
	ONBOARD_EEPROM_SERI_DISABLE	open pins 3 to 4
	ONBOARD_EEPROM_SERI_ENABLE	short pins 3 to 4
	ONBOARD_EEPROM_SERO_DISABLE	open pins 5 to 6
	ONBOARD_EEPROM_SERO_ENABLE	short pins 5 to 6
CS	ONBOARD_EEPROM_CS_ENABLE	short pins 1 to 2
	DIP_EEPROM_CS_ENABLE	short pins 3 to 4

Table 25. Motherboard Output Jumper Configurations

Designator	Value Symbol	Pin Configuration
AOR	AOR_VREG	short pins 1 to 2
	AOR_AGND	short pins 2 to 3
AO-EN	AO0_DISABLE	open pins 1 to 2
	AO1_DISABLE	open pins 3 to 4
	AO0_ENABLE	short pins 1 to 2
	AO1_ENABLE	short pins 3 to 4
RCVR0HP	OD0_HI_PWR_DISABLE	open pins 1 to 2
		open pins 3 to 4
	OD0_HI_PWR_ENABLE	short pins 1 to 2
		short pins 3 to 4
RCVR1HP	OD1_HI_PWR_DISABLE	open pins 1 to 2
		open pins 3 to 4
	OD1_HI_PWR_ENABLE	short pins 1 to 2
		short pins 3 to 4
FILT0EN	OD0_RC_FILT_DISABLE	open pins 1 to 2
		open pins 3 to 4
	OD0_RC_FILT_ENABLE	short pins 1 to 2
		short pins 3 to 4
FILT1EN	OD1_RC_FILT_DISABLE	open pins 1 to 2
		open pins 3 to 4
	OD1_RC_FILT_ENABLE	short pins 1 to 2
		short pins 3 to 4

Table 26. Daughterboard Jumper Configurations

Designator	Value Symbol	Pin Configuration
CP-EN	CHARGEPUMP_INPUT_VBAT_ENABLE	short
	CHARGEPUMP_INPUT_VBAT_DISABLE	open
VH-SEL	VH_CHARGEPUMP_3V3	short pins 1 to 2
	VH_EXTERNAL NOTE: Labels are reversed on the silkscreen	short pins 2 to 3
SW1-SEL	DIO10_ENABLE	short pins 1 to 2
	DIO11_ENABLE	short pins 3 to 4
	DIO12_ENABLE	short pins 5 to 6
	DIO13_ENABLE	short pins 7 to 8
	DIO14_ENABLE	short pins 9 to 10
SW2-SEL	DIO10_ENABLE	short pins 1 to 2
	DIO11_ENABLE	short pins 3 to 4
	DIO12_ENABLE	short pins 5 to 6
	DIO13_ENABLE	short pins 7 to 8
	DIO14_ENABLE	short pins 9 to 10
TRIM-SEL	DIO10_ENABLE	short pins 1 to 2
	DIO11_ENABLE	short pins 3 to 4
	DIO12_ENABLE	short pins 5 to 6
	DIO13_ENABLE	short pins 7 to 8
	DIO14_ENABLE	short pins 9 to 10
PCM-EN	PCM_FRAME_ENABLE	short pins 1 to 2
	PCM_SERI_ENABLE	short pins 3 to 4
	PCM_SERO_ENABLE	short pins 5 to 6
	PCM_CLK_ENABLE	short pins 7 to 8
GPIO-EN	DIO10_ENABLE	short pins 1 to 2
	DIO11_ENABLE	short pins 3 to 4
	DIO12_ENABLE	short pins 5 to 6
	DIO13_ENABLE	short pins 7 to 8
	DIO14_ENABLE	short pins 9 to 10
X1	PCM_LEVEL_TRANSLATOR_ENABLE	short middle pin to "+"
	PCM_LEVEL_TRANSLATOR_DISABLE	short middle pin to "-"
X2	DIO10-13_LEVEL_TRANSLATOR_ENABLE	short middle pin to "+"
	DIO10-13_LEVEL_TRANSLATOR_DISABLE	short middle pin to "-"
X3	DIO14-CLK_LEVEL_TRANSLATOR_ENABLE	short middle pin to "+"
	DIO14-CLK_LEVEL_TRANSLATOR_DISABLE	short middle pin to "-"
X4	LED_ENABLE	short middle pin to "-"
	LED_DISABLE	short middle pin to "+"

A.3 HEADERS

Table 27. Motherboard Header Descriptions

Designator	Description
SDA	I^2C SDA pull-up resistor header
SCL	I^2C SCL pull-up resistor header
I2C	I^2C header
EXT-PSU	External power supply header for main power
EXTCLK	External clock header
VBAT	External VBAT capacitor header
VDDA	External VDDA capacitor header
VBATOD	External VBATOD and VBATOD1 capacitor header
VDDM	External VDDM capacitor header
VDDO1	External VDDO1 capacitor header
VDDO2	External VDDO2 capacitor header
VDDO3	External VDDO3 capacitor header
VDDC	External VDDC capacitor header
VDBL	External VDBL capacitor header
VREGCP	No functionality
VREG	External VREG capacitor header
AI0H	AI0 input header
AI1H	AI1 input header
AI2H	AI2 input header
AI3H	AI3 input header
AI4H	AI4 input header
AI5H	AI5 input header
AO0H	AO0 analog output header
AO1H	AO1 analog output header
FILTEN0	OD0 header, non-filtered output at pins 1 and 3
FILTEN1	OD1 header, non-filtered output at pins 1 and 3
DIO0-9	DIO0 to DIO9 signals
DIO10-19	DIO10 to DIO19 signals
DIO20-29	DIO20 to DIO29 signals

Table 28. Daughterboard Header Descriptions

Designator	Description
CP-EXT	External power supply header for charge pump input voltage
VH-EXT	External power supply header for level translator and LED circuit
PCM-LT	Level translated PCM signals header
GPIO-LT	Level translated GPIO signals header (DIO10-DIO14)
CLK-VH	Level translated clock signal header

A.4 SWITCHES

Table 29. Switch Descriptions

Designator	Description	Board
PSU-SEL	Main power switch select	Motherboard
SW1	Momentary pushbutton switch for selected DIO	Daughterboard
SW2	Momentary pushbutton switch for selected DIO	Daughterboard

Table 30. Switch Configurations

Designator	Value Symbol	Pin Configuration
PSU-SEL	MAIN_POWER_CAA	Switch to CAA
	MAIN_POWER_EXT	Switch to EXT
SW1	DIO_SEL_SW1_HIGH	Switch open
	DIO_SEL_SW1_LOW	Switch closed
SW2	DIO_SEL_SW2_HIGH	Switch open
	DIO_SEL_SW2_LOW	Switch closed

A.5 CONNECTORS

Table 31. Motherboard Connector Descriptions

Designator	Description
J1	6-pin DIN connector to Communication Accelerator Adaptor or other Ezairo 7100 development board; supports Vsense
J2	6-pin DIN connector to Communication Accelerator Adaptor or other Ezairo 7100 development board
DIO5-9	10-pin protected header to Ezairo 7100 DIO5 to DIO9 signals
DIO25-29	10-pin protected header to Ezairo 7100 DIO25 to DIO29 signals
AI0	AI0 input RCA jack
AI1	AI1 input RCA jack
AI2	AI2 input RCA jack
AI3	AI3 input RCA jack
AI4	AI4 input RCA jack
AI5	AI5 input RCA jack
RCVR0	OD0 direct digital output RCA jack
RCVR1	OD1 direct digital output RCA jack
AO0	AO0 output RCA jack
AO1	AO1 output RCA jack
J3	15-pin DSUB connector to any Ezairo series daughterboard

Table 32. Daughterboard Connector Descriptions

Designator	Description
J1	15-pin DSUB connector to any Ezairo series motherboard

A.6 POTENTIOMETERS

Table 33. Daughterboard Potentiometer Description

Designator	Description
R2	LSAD volume control

CHAPTER B
Motherboard Schematics

This appendix contains all schematics for the Evaluation and Development Board motherboard:

- Communication Interface and Power Supply Schematic
- DIOs and EEPROM Schematic
- Ezairo 7100 Schematic

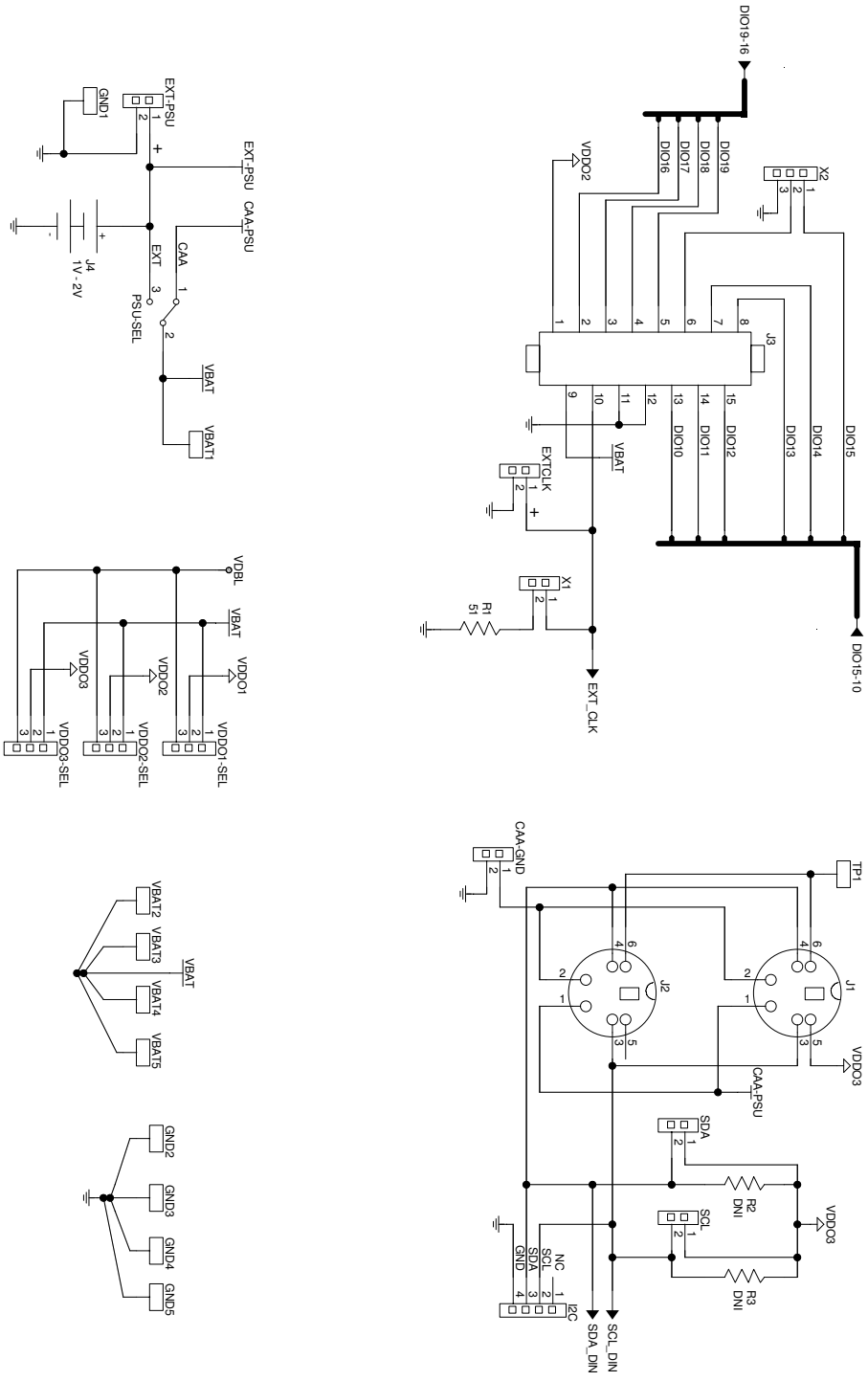


Figure 4. Motherboard - Communication Interface and Power Supply Schematic

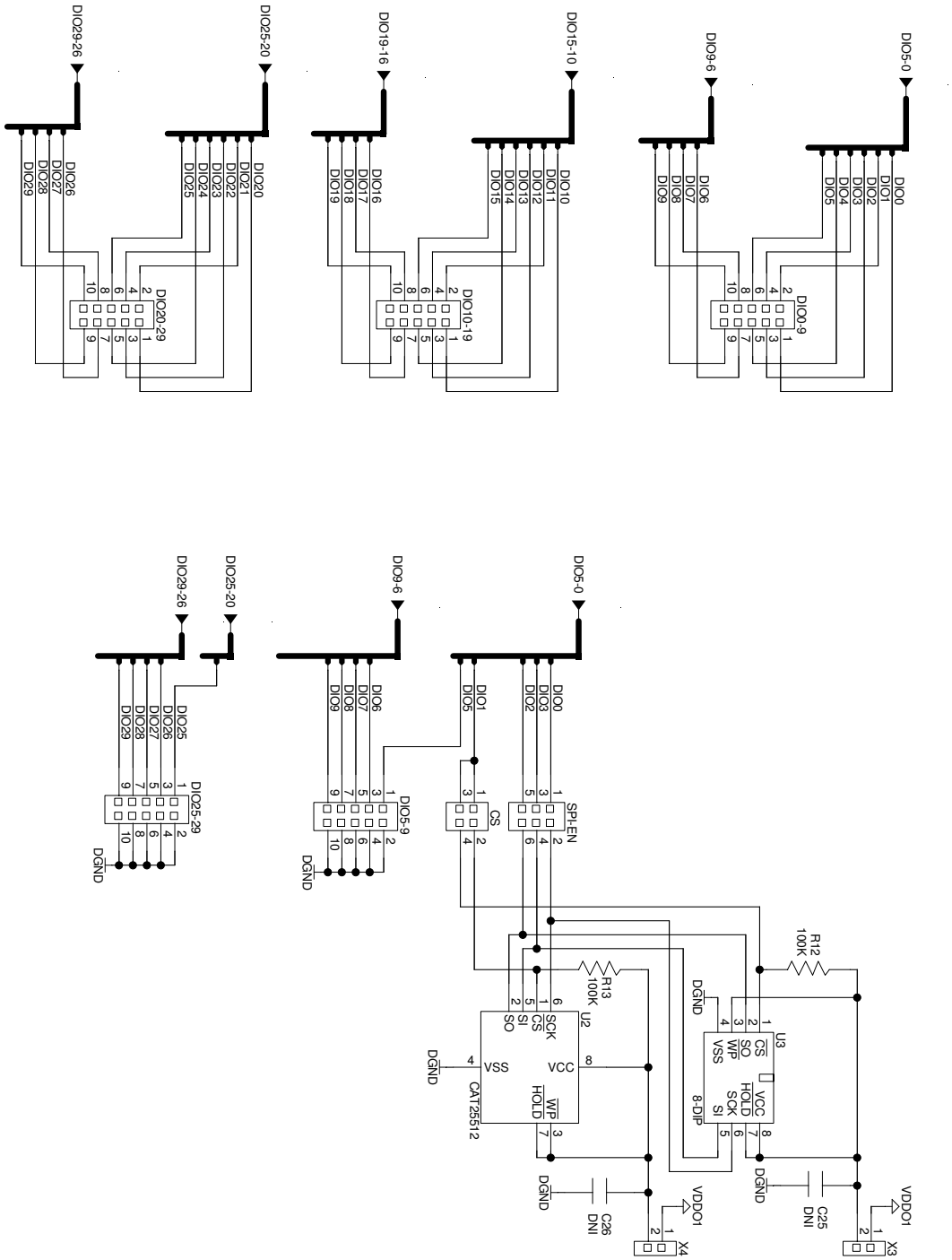


Figure 5. Motherboard - DIOs and EEPROM Schematic

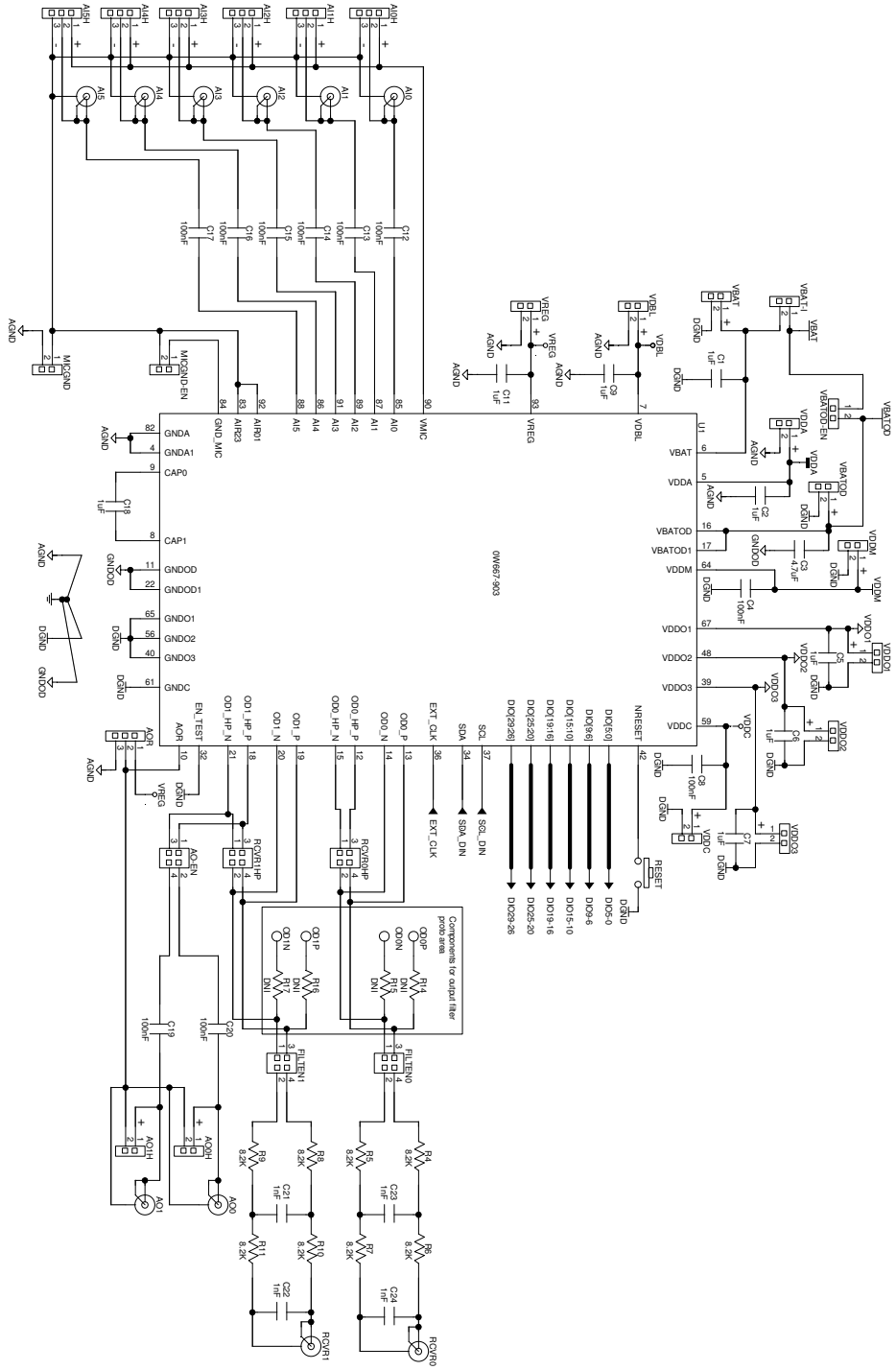


Figure 6. Motherboard - Ezairo 7100 Schematic

CHAPTER C
Daughterboard Schematics

This appendix contains all schematics for the Evaluation and Development Board daughterboard:

- PSU/DIO-LSAD Interface Schematic
- Level Translator Schematic

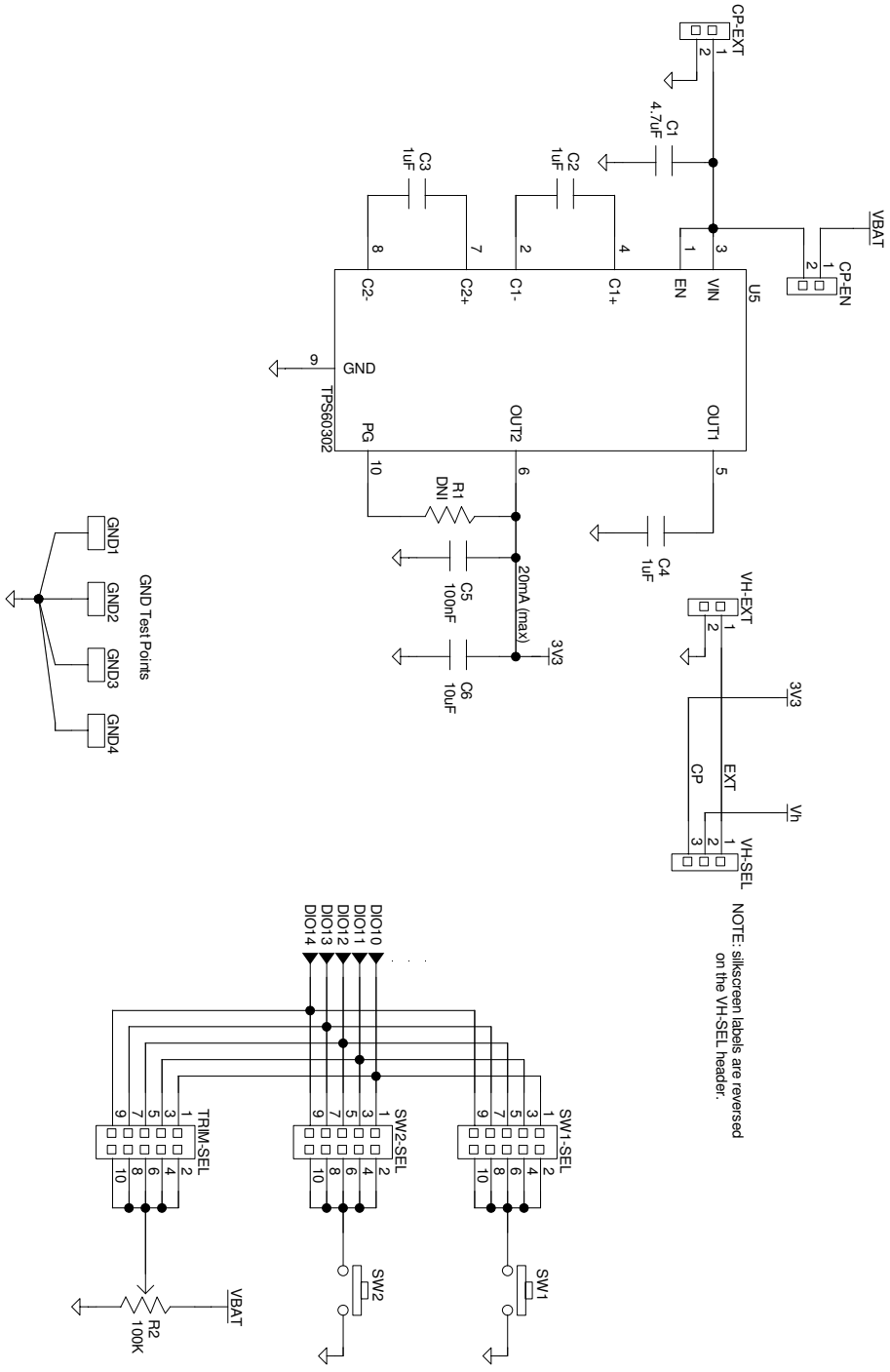


Figure 7. Daughterboard - PSU/DIO-LSAD Interface Schematic

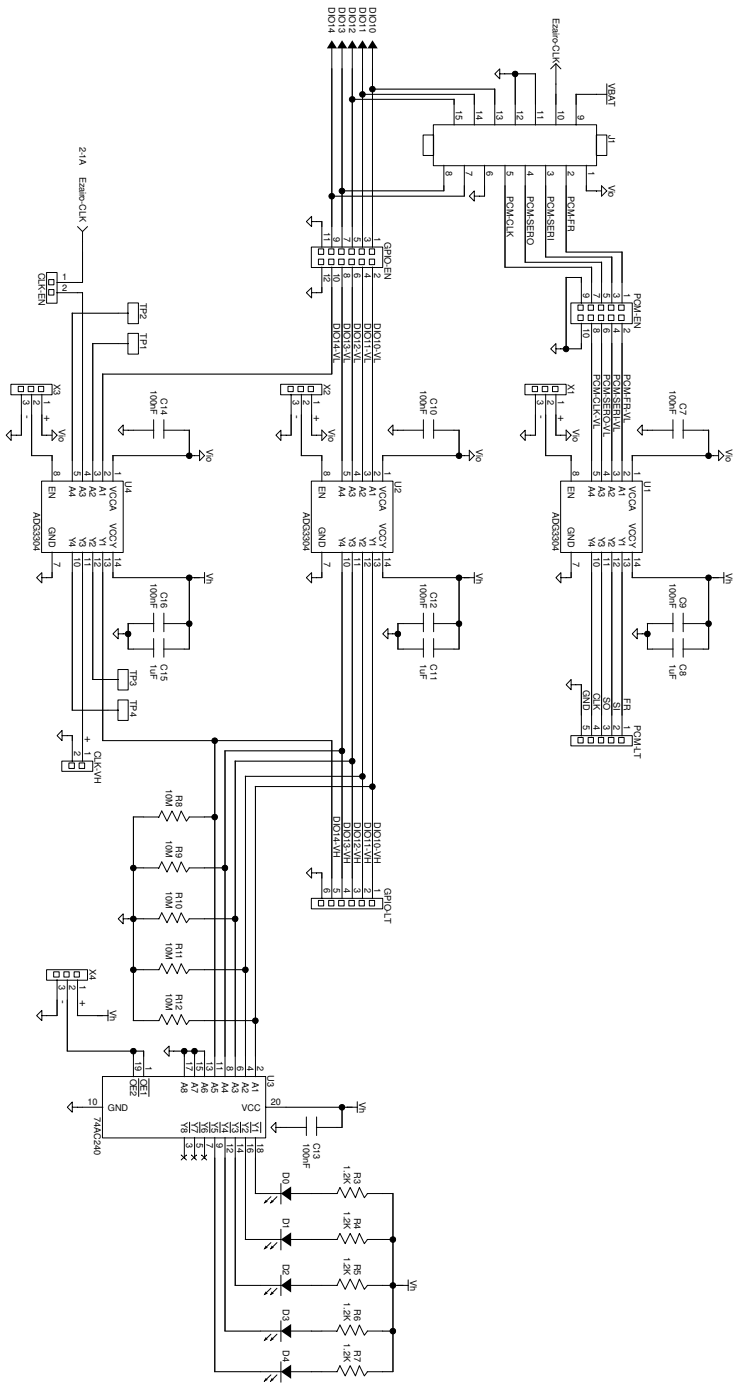


Figure 8. Daughterboard - Level Translator Schematic

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